Line Drawing

Primary Algorithms:

- **Digital Differential Analyzer (DDA)**: The DDA algorithm is thoroughly developed as a scan-conversion method that samples at unit intervals in one coordinate and computes corresponding integer values for the other. The two cases based on slope magnitude:
  - For $|m| < 1$: sampling at unit x intervals with y incremented by m each step
  - For $|m| > 1$: sampling at unit y intervals with x incremented by 1/m each step
  - The algorithm advantages: simplicity, speed relative to direct equation evaluation
  - The algorithm disadvantages: floating-point arithmetic overhead, round-off error accumulation causing pixel drift
- **Bresenham's Midpoint Algorithm**: it's an accurate and efficient alternative that uses only incremental integer calculations.
  - For lines with positive slope less than one where $|\Delta x| > |\Delta y|$, the algorithm makes decisions about the next pixel position (xk+1, yk) or (xk+1, yk+1) based on a decision parameter pk. The complete algorithm is specified with initialization ($p0 = 2\Delta y - \Delta x$) and update rules:
    - If $pk < 0$: Setpixel (xk+1, yk), $pk+1 = pk + 2\Delta y$
    - Otherwise: Setpixel (xk+1, yk+1), $pk+1 = pk + 2\Delta y - 2\Delta x$
  - For lines with positive slope greater than one where $|\Delta y| > |\Delta x|$ (requiring coordinate swapping),
  - For negative slopes (coordinate decrements)
  - The algorithm advantages: it's significantly faster due to integer-only operations; also it's more accurate and efficient, without round-off errors.
  - The algorithm disadvantages: A full implementation must handle all combinations of slope and endpoint order by adjusting step sizes and interchanging X and Y coordinates, adding complexity.

Solve the following questions

- **Use the DDA algorithm to draw a line from A(2, 0) to B(6, 3). Show all intermediate pixel positions.**

Answer:

$|\Delta x|=|6-2|=4$, $|\Delta y|=|3-0|=3 \rightarrow |\Delta x| > |\Delta y|$

$xk+1=xk+1$, $yk+1=yk+m$

$m=0.75$

x	y	Setpixel(x,y)
2	0	Setpixel(2,0)
3	0.75	Setpixel(3,1)
4	1.5	Setpixel(4,2)
5	2.25	Setpixel(5,2)
6	3	Setpixel(6,3)

- **Draw a line from (3, 2) to (7, 8) using DDA algorithm.**

Answer:

|Δx|=|7-3|=4, |Δy|=|8-2|=6 → |Δy|> |Δx|

xk+1=xk+1/m , yk+1=yk+1

1/m=0.667

x	y	Setpixel(x,y)
3	2	Setpixel(3,2)
3.667	3	Setpixel(4,3)
4.33	4	Setpixel(4,4)
5	5	Setpixel(5,5)
5.66	6	Setpixel(6,6)
6.33	7	Setpixel(6,7)
7	8	Setpixel(7,8)

- **Using Bresenham's algorithm, digitize the line with endpoints (20, 2) and (12, 9).**

Answer:

|Δx|=|12-20|=|-8|, |Δy|=|9-2|=|7| → |Δx|> |Δy|

p0 = 2Δy – Δx=14-8=6

- If pk < 0: Setpixel (xk-1, yk), pk+1 = pk + 14
- Otherwise: Setpixel (xk-1, yk+1), pk+1 = pk -2

P	x	y	Setpixel(x,y)
-	20	2	Setpixel(20,2)
6	19	3	Setpixel(19,3)
4	18	4	Setpixel(18,4)
2	17	5	Setpixel(17,5)
0	16	6	Setpixel(16,6)
-2	15	6	Setpixel(15,6)
12	14	7	Setpixel(14,7)
10	13	8	Setpixel(13,8)
8	12	9	Setpixel(12,9)

- **Using Bresenham's algorithm, digitize the line with endpoints (7, 7) and (11, 13).**

Answer:

|Δx|=|11-7|=|4|, |Δy|=|13-7|=|6| → |Δy|> |Δx|

Swap x points with y points
X0=7, xend=13 & y0=7, yend=11
So
|Δx|=|13-7|=6, |Δy|=|11-7|=4 p0 = 2Δy – Δx= 2

- If pk < 0: Setpixel (yk+1, xk), pk+1 = pk + 8
- Otherwise: Setpixel (yk+1, xk+1), pk+1 = pk -4

P	X	y	**Setpixel(y,x)**
-	7	7	Setpixel(7,7)
2	8	8	Setpixel(8,8)
-2	9	8	Setpixel(8,9)
6	10	9	Setpixel(9,10)
2	11	10	Setpixel(10,11)
-2	12	10	Setpixel(10,12)
6	13	11	Setpixel(11,13)